

Partial Progress on the Li–Goldberg Covariance Conjecture for In-Service Work in an $M/GI/n$ Queue

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Abstract

Li and Goldberg conjectured that, in a stationary first-come-first-served $M/GI/n$ queue, the number of busy servers and the total residual work in service have nonnegative covariance. This note gives a corrected and self-contained version of a short attempted proof. The unrestricted conjecture is *not* proved here. The contribution is partial progress: we prove exact covariance identities, identify the conjecture with a sharp mean-queue-length inequality, give an exact finite-dimensional conditional-mean criterion, prove the conjecture in the low offered-load regime $\lambda \mathbb{E} S \leq 1$, and record two standard solved cases: $M/GI/1$ and $M/M/n$. We also separate two useful comparison systems, $M/GI/\infty$ and $M/GI/n/n$, which support the intuition but are not instances of the original waiting-line conjecture.

Status. This is a partial-progress note, not a solution of the Li–Goldberg conjecture for all stable FCFS $M/GI/n$ queues. In particular, the positive association and conditional-monotonicity statements below are sufficient conditions unless explicitly labelled otherwise.

1 Model and notation

Consider a stable stationary FCFS $M/GI/n$ queue with Poisson arrival rate λ , i.i.d. service requirements S satisfying $0 < \mathbb{E} S < \infty$ and $\mathbb{E} S^2 < \infty$, and traffic intensity

$$\rho = \frac{\lambda \mathbb{E} S}{n} < 1.$$

At a stationary time, let

$$B = \text{Num}_{\text{service}}$$

be the number of busy servers, let

$$W_s = \text{Work}_{\text{service}}$$

be the sum of the residual service times of jobs currently in service, let Q be the number of jobs waiting in queue, and let W_q be the sum of the service requirements of the jobs waiting in queue. The total unfinished work in the system is

$$V = W_s + W_q.$$

The conjectured inequality is

$$\mathbb{E} B \mathbb{E} W_s \leq \mathbb{E}[BW_s], \quad \text{equivalently} \quad \text{Cov}(B, W_s) \geq 0. \quad (1)$$

All expectations appearing below are assumed finite. This assumption is made only to avoid technical distractions in applying stationary drift identities; one may alternatively work with the usual truncation/localization arguments.

2 Exact identities

The next proposition is the main algebraic reduction. It is essentially the same correction-term phenomenon that appears in Li–Goldberg’s Theorem 9, but we include the proof because it is short and clarifies exactly what remains to be shown.

Proposition 2.1 (Covariance–queue-length identity). *In the stationary FCFS $M/GI/n$ queue described above,*

$$\mathbb{E} B = \lambda \mathbb{E} S, \quad (2)$$

$$\mathbb{E} W_s = \frac{\lambda \mathbb{E} S^2}{2}, \quad (3)$$

$$\mathbb{E} W_q = (\mathbb{E} S) \mathbb{E} Q, \quad (4)$$

$$\text{Cov}(B, V) = \frac{\lambda \mathbb{E} S^2}{2}, \quad (5)$$

$$\text{Cov}(B, W_s) = \frac{\lambda \mathbb{E} S^2}{2} - n(1 - \rho)(\mathbb{E} S) \mathbb{E} Q. \quad (6)$$

Consequently, (1) is equivalent to the sharp mean-queue-length bound

$$\mathbb{E} Q \leq \frac{\lambda \mathbb{E} S^2}{2n(1 - \rho) \mathbb{E} S} = \frac{1}{2} \frac{\mathbb{E} S^2}{(\mathbb{E} S)^2} \frac{\rho}{1 - \rho}. \quad (7)$$

Proof. Equation (2) is the stationary rate-conservation identity for total busy time. Over a long interval, the time integral of $B(t)$ differs from the total service requirement of jobs completed in that interval only by boundary terms; after division by the interval length and passage to stationarity this gives $\mathbb{E} B = \lambda \mathbb{E} S$.

For (3), a job with service requirement s contributes

$$\int_0^s (s - u) du = \frac{s^2}{2}$$

to the time integral of in-service residual work. In a stable lossless queue the service-start rate equals the arrival rate λ , so the renewal-reward identity yields $\mathbb{E} W_s = \lambda \mathbb{E} S^2 / 2$.

For (4), use the standard deferred-decision construction: the service requirement of a waiting job need not be revealed until the job reaches service, since FCFS does not use it before service begins. Conditional on Q , the Q unrevealed waiting requirements are therefore i.i.d. copies of S , giving $\mathbb{E}[W_q | Q] = (\mathbb{E} S)Q$.

Now consider the total unfinished work V . Between arrivals, V decreases at instantaneous rate B . At an arrival, V jumps upward by an independent copy of S . Applying the stationary drift identity to the quadratic test function $v \mapsto v^2$ gives

$$0 = -2 \mathbb{E}[BV] + \lambda \mathbb{E}[(V + S)^2 - V^2] = -2 \mathbb{E}[BV] + 2\lambda(\mathbb{E} S) \mathbb{E} V + \lambda \mathbb{E} S^2.$$

Together with (2), this is exactly

$$\mathbb{E}[BV] - (\mathbb{E} B)(\mathbb{E} V) = \frac{\lambda \mathbb{E} S^2}{2},$$

which proves (5). Finally, whenever $W_q > 0$ all n servers are busy, and the identity $BW_q = nW_q$ also holds when $W_q = 0$. Thus

$$\text{Cov}(B, W_q) = n \mathbb{E} W_q - (\mathbb{E} B)(\mathbb{E} W_q) = n(1 - \rho) \mathbb{E} W_q.$$

Since $V = W_s + W_q$, equations (4) and (5) imply (6). Rearranging (6) gives (7). $\square$

Remark 2.2. *Proposition 2.1 does not solve the conjecture. It restates the covariance problem as the mean-queue-length bound (7). A proof of (7) for every stable FCFS $M/GI/n$ queue would be equivalent to a proof of the conjecture.*

3 Immediate consequences

3.1 A genuine low-load regime

The following elementary observation proves the conjecture whenever the total offered load is at most one job of work, regardless of the number of servers. For $n = 1$ this is the whole stability region; for $n \geq 2$ it gives the regime $\rho \leq 1/n$.

Theorem 3.1 (Low offered load). *If $\lambda \mathbb{E} S \leq 1$, then*

$$\text{Cov}(B, W_s) \geq 0.$$

More precisely,

$$\text{Cov}(B, W_s) \geq \frac{\lambda \mathbb{E} S^2}{2} (1 - \lambda \mathbb{E} S). \quad (8)$$

Proof. Pointwise, $BW_s \geq W_s$, because $W_s > 0$ implies that at least one server is busy. Hence

$$\text{Cov}(B, W_s) = \mathbb{E}[BW_s] - (\mathbb{E} B)(\mathbb{E} W_s) \geq \mathbb{E} W_s (1 - \mathbb{E} B).$$

Using (2) and (3) proves (8). The right-hand side is nonnegative if $\lambda \mathbb{E} S \leq 1$. □

Corollary 3.2 (Single server). *For every stable stationary $M/GI/1$ queue with $\mathbb{E} S^2 < \infty$,*

$$\text{Cov}(B, W_s) = (1 - \rho) \mathbb{E} W_s \geq 0.$$

Proof. When $n = 1$, $B = \mathbf{1}\{W_s > 0\}$ and therefore $BW_s = W_s$. Thus

$$\text{Cov}(B, W_s) = \mathbb{E} W_s - (\mathbb{E} B)(\mathbb{E} W_s) = (1 - \rho) \mathbb{E} W_s.$$

□

3.2 Exact conditional-mean criterion

Let

$$p_k = \mathbb{P}(B = k), \quad m_k = \mathbb{E}[W_s \mid B = k], \quad k = 0, 1, \dots, n,$$

with arbitrary values assigned to m_k if $p_k = 0$. Since $W_s = 0$ on $\{B = 0\}$, one may take $m_0 = 0$.

Proposition 3.3 (Finite conditional-mean profile). *The covariance has the exact representation*

$$\text{Cov}(B, W_s) = \sum_{0 \leq i < j \leq n} p_i p_j (j - i) (m_j - m_i). \quad (9)$$

Consequently, the conjecture is equivalent to the nonnegativity of the weighted sum in (9). In particular, it is sufficient, but not necessary, that $k \mapsto m_k$ be nondecreasing on the support of B .

Proof. By conditioning on B ,

$$\text{Cov}(B, W_s) = \text{Cov}(B, \mathbb{E}[W_s | B]) = \text{Cov}(B, m_B).$$

If X' is an independent copy of a real random variable X , then

$$\text{Cov}(X, h(X)) = \frac{1}{2} \mathbb{E}[(X - X')(h(X) - h(X'))].$$

Applying this identity to $X = B$ and $h(k) = m_k$ gives (9) after summing over the pairs $i < j$. The final assertion follows because all weights $p_i p_j (j - i)$ are nonnegative. $\square$

Corollary 3.4 (Two-server profile). *For $n = 2$, the conjecture is equivalent to*

$$p_0 p_1 m_1 + 2p_0 p_2 m_2 + p_1 p_2 (m_2 - m_1) \geq 0.$$

In particular, a possible failure for two servers would require the conditional mean in-service work at two busy servers to be sufficiently smaller than at one busy server; mere nonmonotonicity would not by itself be enough.

4 Positive-dependence sufficient conditions

The next two results are useful because they isolate exactly what kind of positive dependence would settle the conjecture. They should not be read as proofs that the unrestricted FCFS system has the required dependence.

Definition 4.1 (Association). *A random vector $X = (X_1, \dots, X_m)$ is associated if*

$$\text{Cov}(f(X), g(X)) \geq 0$$

for all bounded coordinatewise nondecreasing Borel functions f, g for which the covariance exists.

Proposition 4.2 (Association of residuals is sufficient). *Let $R = (R_1, \dots, R_n)$ be the stationary vector of residual service times at the servers, with $R_i = 0$ if server i is idle. If R is associated, then*

$$\text{Cov}\left(\sum_{i=1}^n \mathbf{1}\{R_i > 0\}, \sum_{i=1}^n R_i\right) \geq 0.$$

Thus the Li–Goldberg covariance inequality holds.

Proof. The maps

$$r \mapsto \sum_{i=1}^n \mathbf{1}\{r_i > 0\}, \quad r \mapsto \sum_{i=1}^n r_i$$

are coordinatewise nondecreasing on $\mathbb{R}_+^n$. Apply association to bounded truncations of the second map and then pass to the limit by monotone convergence. $\square$

Proposition 4.3 (Monotone conditional mean is sufficient). *If*

$$m_k = \mathbb{E}[W_s | B = k]$$

is nondecreasing in k , then $\text{Cov}(B, W_s) \geq 0$.

Proof. This is the immediate sufficient condition in Proposition 3.3. Equivalently,

$$\text{Cov}(B, W_s) = \text{Cov}(B, m_B) = \frac{1}{2} \mathbb{E} [(B - B')(m_B - m_{B'})] \geq 0,$$

where B' is an independent copy of B . □

Remark 4.4. *The monotonicity and association hypotheses are stronger than the conjecture. The exact criterion is (9); monotonicity merely guarantees that every summand in that criterion is nonnegative.*

5 A standard solved case: exponential service

Theorem 5.1 ($M/M/n$). *For every stable FCFS $M/M/n$ queue with exponential service rate μ ,*

$$\text{Cov}(B, W_s) = \frac{1}{\mu} \text{Var}(B) \geq 0.$$

Proof. By the memoryless property, conditional on $B = k$, the k in-service residual service times are i.i.d. exponential random variables with mean $1/\mu$. Hence

$$\mathbb{E}[W_s \mid B = k] = \frac{k}{\mu}.$$

Therefore

$$\text{Cov}(B, W_s) = \text{Cov}(B, \mathbb{E}[W_s \mid B]) = \text{Cov}\left(B, \frac{B}{\mu}\right) = \frac{1}{\mu} \text{Var}(B) \geq 0.$$

□

6 Comparison systems outside the waiting-line conjecture

The systems in this section are not counterexamples and not proofs of the finite- n FCFS waiting-line conjecture. They are included only to clarify which parts of the attempted proof were valid but off-model.

Proposition 6.1 ($M/GI/\infty$). *In a stationary $M/GI/\infty$ system,*

$$\text{Cov}(B, W_s) = \mathbb{E} W_s = \frac{\lambda \mathbb{E} S^2}{2} \geq 0.$$

Proof. At stationarity, active jobs form a Poisson random measure on age-service pairs (a, s) with intensity

$$\lambda \mathbf{1}\{0 < a < s\} da dF_S(s).$$

For a Poisson random measure N with intensity measure ν ,

$$\text{Cov}\left(\int f dN, \int g dN\right) = \int fg d\nu.$$

Take $f(a, s) = 1$ and $g(a, s) = s - a$. Then

$$\text{Cov}(B, W_s) = \lambda \int_0^\infty \int_0^s (s - a) da dF_S(s) = \frac{\lambda \mathbb{E} S^2}{2}.$$

□

Proposition 6.2 ($M/GI/n/n$ loss system). *In the stationary Erlang loss system $M/GI/n/n$, assuming the standard insensitive product-form stationary description, one has*

$$\text{Cov}(B, W_s) = r_S \text{Var}(B) \geq 0, \quad r_S = \frac{\mathbb{E} S^2}{2 \mathbb{E} S}.$$

Proof. The product-form description of the Erlang loss system says that, conditional on $B = k$, the k in-service residual service times are i.i.d. with the stationary-excess distribution of S , whose mean is $r_S = \mathbb{E} S^2 / (2 \mathbb{E} S)$. Hence

$$\mathbb{E}[W_s \mid B = k] = k r_S.$$

The same conditional-mean calculation as in Theorem 5.1 gives $\text{Cov}(B, W_s) = r_S \text{Var}(B) \geq 0$. $\square$

7 What remains open

The exact identity (6) shows that a full proof of the Li–Goldberg conjecture is equivalent to proving

$$\mathbb{E} Q \leq \frac{1}{2} \frac{\mathbb{E} S^2}{(\mathbb{E} S)^2} \frac{\rho}{1 - \rho}$$

for every stable FCFS $M/GI/n$ queue for which the steady-state quantities are well-defined. The results above prove this in the low offered-load region $\lambda \mathbb{E} S \leq 1$, for $n = 1$, and for exponential service. Outside these cases, the missing ingredient is a genuine in-service residual-work statement, such as the weighted conditional-mean criterion (9), not merely a statement about the total workload.

A tempting Kiefer–Wolfowitz workload-vector argument is insufficient by itself: the ordered workload vector naturally controls total unfinished work, including waiting jobs already assigned to future service, whereas the conjecture concerns only residual work currently in service. The difference is exactly the term $n(1 - \rho)(\mathbb{E} S) \mathbb{E} Q$ in (6).

References

- [1] Yuan Li and David A. Goldberg. *Simple and explicit bounds for multi-server queues with $1/(1 - \rho)$ scaling*. *Mathematics of Operations Research* 50(2), 813–837, 2025. arXiv:1706.04628, DOI: [10.1287/moor.2022.0131](https://doi.org/10.1287/moor.2022.0131).
- [2] Frank P. Kelly. *Reversibility and Stochastic Networks*. Wiley, 1979.
- [3] Pranav Nuti, Eric Fithian, and Rad Niazadeh. *Open Problems in Operations Research*, problem W4393423947_p0. https://pranav-nuti.github.io/open-problems-in-or/problem.html?id=W4393423947_p0&page=11&n=116.